

Dawar Zaman

MECHANICAL ENGINEER | PEC REGISTERED
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Profile

Safety-focused Mechanical Engineer with hands-on technical ability, wellsite awareness, and strong safety mindset. Certified in **IWCF Level 1** Well Control and **Well Operations Crew Resource Management**, demonstrating understanding of safe communication, situational awareness, and field teamwork. Skilled in mechanical analysis, CAD modeling, simulations, and troubleshooting. Committed to learning, discipline, and contributing effectively in demanding operational environments.

Core Skills

- **Technical & Engineering Software:** SolidWorks | AutoCAD | ANSYS (FEA/Structural) | COMSOL | MATLAB | Python | C++
- **Oil & Gas Operations:** Well Control Awareness | BOP Stack | Cementing Units Coiled Tubing | Rig Up/Down | Operational Reporting
- **Engineering Analysis:** FEA | Structural & Thermal Analysis | Mechanics of Materials | Design Validation | Prototyping
- **Safety & Compliance:** IWCF Level 1 | Job Safety Analysis (JSA) | Hazard Identification | Safe Work Practices | HSE Culture | Crew Resource Management (CRM) Principles
- **Simulation & FEA:** ANSYS | COMSOL – structural & thermal analysis
- **Programming & Computation:** MATLAB | Python | C++ – data analysis, ANN, GA optimization
- **Technical Reporting:** MS Word | Excel – Clear documentation, results analysis, and engineering communication

Soft Skills

- Problem Solving | Project Management | Team Collaboration | Field Communication | Adaptability | Time Management | Attention to Detail | Critical Thinking | Working under pressure | Results oriented | Situational Awareness

Experience

BGP International (Oil and Gas) | HSE Intern

Dec 2025

- Supported implementation and monitoring of **HSE Management Systems** in compliance with **oil & gas safety standards and company HSE policies**.
- Conducted **site safety inspections, hazard identification, and risk assessments**, contributing to safer field operations.
- Assisted with **Job Safety Analysis (JSA)** and **Permit to Work (PTW)** processes to ensure controlled and compliant work activities.
- Participated in **incident, near-miss reporting, HSE documentation, and PPE compliance monitoring** to strengthen workplace safety culture.

Heavy Industries Taxila | Intern

Jul 2024-Aug 2024 (2 mos.)

- **Applied FEA and CAD** modeling to validate the structural integrity of a barrel integration on an Armored Personnel Carrier (APC).
- **Modeled and simulated** assembly designs in SolidWorks to test performance under combat conditions.
- Performed design validation using engineering simulations (**FEA/structural analysis**) to ensure safety, **performance**, and durability under combat conditions.
- **Collaborated** with senior engineers to refine designs, contributing to a critical **defense project**.

Projects

FYP: AI Based optimization of Piezo-electric Energy Harvesters

- Designed a **3D model** of a **cantilever-type** piezoelectric harvester array using **Solid Works**.
- Generated 109 geometric combinations and simulated every configuration on **COMSOL**.
- Used a Genetic Algorithm (**GA**) on the trained ANN model to determine **optimal design** for **maximum energy generation**.
- Attained a substantial **power output gain** over the reference study.

Mechanics of Materials Project: Design & Analysis of a transformer platform.

- Designed a **steel platform** to carry a 1500 kVA, **4000 kg transformer**.
- Checked **beam & column safety** based on **principles of mechanics**.
- Modeled structural performance with ANSYS Structural.
- Verified cost-efficient and stable design by analysis and iteration.

Sem-Project: Design and Structural Analysis of a University Bike Parking Rack

- Designed a campus **bike parking rack** using **statics** and **mechanics of materials** principles.
Developed a detailed **3D CAD** model with assemblies in SolidWorks.
- Performed **structural analysis** and simulated failure modes in ANSYS
- Validated design **for strength, load-bearing capacity**, and real-world functionality

Education

BS Mechanical Engineering | Air University Islamabad

2021-25

Major: Mechanical Engineering

FSC | Fazaia Inter College Minhas

2020-21

Major: Pre-Engineering

Certifications

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- **IELTS Academic** – 7 bands
 - **IWCF Level 1** – Well Control Awareness
 - **WOCRM** – Well Operations Crew Resource Management
 - **ISO 45001:2018** - Principles of Occupational Health and Safety Management Systems
 - **Duke University** : Oil & Gas Industry Operations and Markets

Languages

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- **English** – Fluent (Professional Working Proficiency)
 - **Urdu** – Native / Fluent
 - **German** – Beginner | A1